

2D Array Practice Problems

1. Write a program to input a 2D array of size $M \times N$ (M rows and N columns) and display it in a tabular format, where each element is printed in a grid-like structure with proper alignment.
2. Write a program to input a 2D array of size $M \times N$ and calculate the sum and average of all its elements. Print both the sum and the average as the output. For example, if the matrix is $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the sum would be 10 and the average would be 2.5.
3. Write a program to input a 2D array of size $M \times N$ and display the sum of each row and each column separately. For example, for the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the row sums would be $[3, 7]$ and the column sums would be $[4, 6]$.
4. Write a program to check whether a given 2D array is a transpose matrix. A matrix is said to be a transpose of itself if element at position $[i][j]$ is the same as the element at position $[j][i]$. For example, the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ is its own transpose, while $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is not.
5. Write a program to input a square matrix and check whether it is an identity matrix or not. An identity matrix has 1s on the diagonal from the top-left to bottom-right and 0s elsewhere. For example,

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,

$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,

$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ is an identity matrix.

6. Write a program to input a matrix of order $M \times N$ and check if it is a sparse matrix or a dense matrix. A sparse matrix has most of its elements as zero (generally more than half), whereas a dense matrix has most elements as non-zero. Print whether the matrix is sparse or dense.
7. Write a program to check whether a square matrix is a lower triangular matrix or not. A lower triangular matrix is a square matrix where all elements above the main diagonal are zero. Example: The matrix

$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$,

$\begin{bmatrix} 2 & 3 & 0 \\ 4 & 5 & 6 \\ 0 & 0 & 0 \end{bmatrix}$,

$\begin{bmatrix} 4 & 5 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is a lower triangular matrix, while

$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 6 \end{bmatrix}$,

$\begin{bmatrix} 0 & 3 & 4 \\ 1 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix}$,

$\begin{bmatrix} 0 & 0 & 6 \\ 1 & 2 & 0 \\ 0 & 3 & 4 \end{bmatrix}$ is not.

8. Write a program in C to calculate the determinant of a 3×3 matrix using standard determinant formula. For example, the determinant of matrix

[[a, b, c],

[d, e, f],

[g, h, i]] is calculated as $a(ei - fh) - b(di - fg) + c(dh - eg)$.

9. Write a program to keep records and perform statistical analysis for a class of up to 10 students. Each student will have three quiz scores and be identified by a unique four-digit roll number. The program should print the quiz scores of each student and perform the following tasks:
 - a) Print the highest and lowest score for each quiz along with the roll number of the respective students.
 - b) Calculate and display the average score for each quiz and the overall average of all quizzes.
 - c) Identify the quiz with the highest and lowest average scores.
 - d) Print the highest and lowest marks of each student along with the quiz number.
10. Suppose there is a game called "MATCH THE TABLES" where the player picks two tables, each with 10 rows and 10 columns, and tries to match them. If at least 90 corresponding entries out of the total 100 match between the two tables, the tables are considered identical, and the player is declared the winner. Write a program to simulate this game and determine whether the player wins or not.